

ME 333
MECHANICS OF MATERIALS

Lecture 6: Concept of Stress

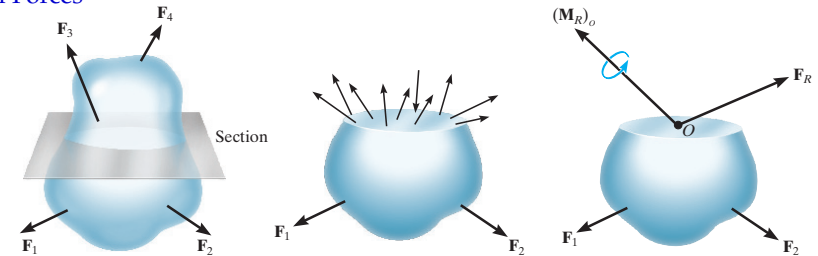
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Sem I, 2025-2026

Concept of Stress

Internal Forces

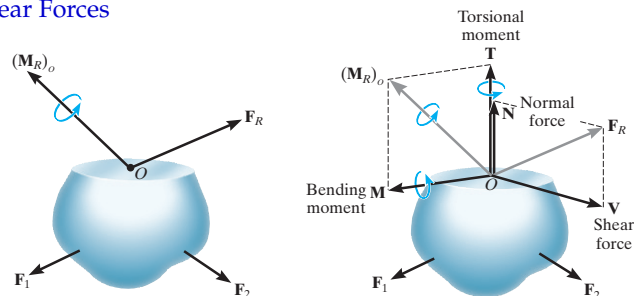


- Internal forces at any arbitrary point O in the body
- F_R is the resultant internal force acting at point O
- $(M_R)_O$ is the resultant internal moment about O

Ref Base Figs: Statics & Mechanics of Materials, R. C. Hibbeler

Concept of Stress

Normal and Shear Forces

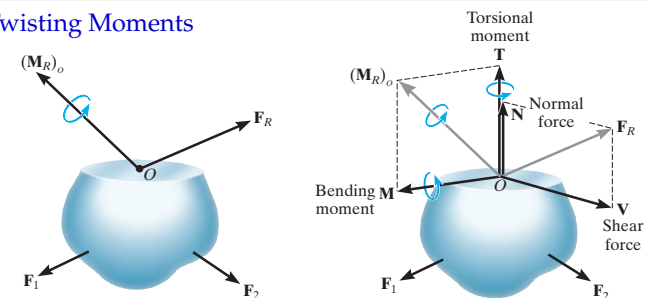


- Force that acts perpendicular to the area is called *Normal Force*
- Force that lies in the plane of the area is called *Shear Force*

Ref Base Figs: Statics & Mechanics of Materials, R. C. Hibbeler

Concept of Stress

Bending and Twisting Moments

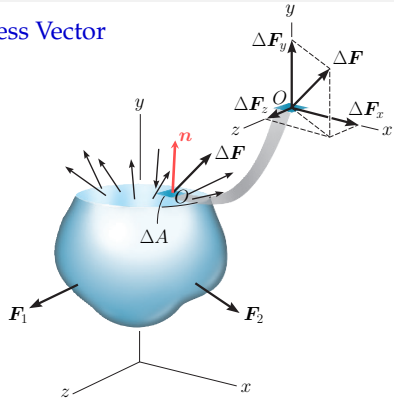


- Moment acting about an axis lying in the plane of the cross-sectional area is called *Bending Moment*
- Moment acting about an axis perpendicular to the cross-sectional area is called *Twisting Moment*

Ref Base Figs: Statics & Mechanics of Materials, R. C. Hibbeler

Concept of Stress

Stress Vector



- Stress Vector $\overset{(n)}{T}$ at point O is:

$$\overset{(n)}{T} = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A}$$

Also, known as *Traction* vector or *force intensity*

- In terms of rectangular components:

$$\overset{(n)}{T} = T_x \hat{i} + T_y \hat{j} + T_z \hat{k}$$

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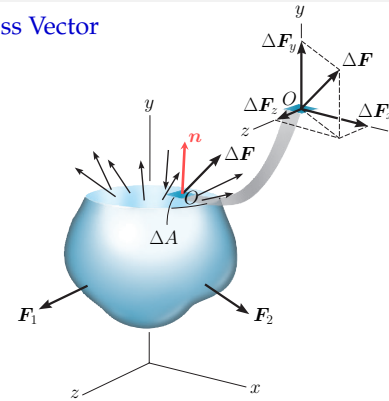
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Concept of Stress

Stress Vector



Characteristics:

- dimensions of force per unit area
- defined at a point on an imaginary plane that divides the material into two
- stress is a vector equivalent to the action of one part of material on the other
- direction of stress vector is *arbitrary*

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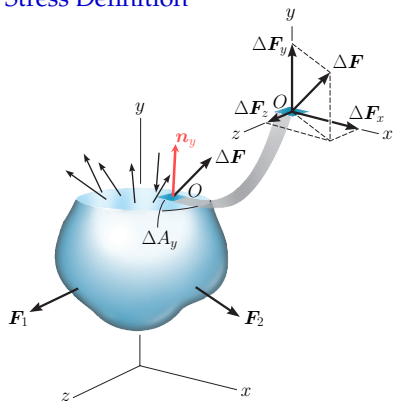
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Concept of Stress

Stress Definition



- Stress vector at point O is :

$$\overset{(n_y)}{T} = \lim_{\Delta A_y \rightarrow 0} \frac{\Delta F}{\Delta A_y}$$

- Three component of stress vector on face n_y at O is:

$$\overset{(n_y)}{T}_x = \lim_{\Delta A_y \rightarrow 0} \frac{\Delta F_x}{\Delta A_y} = \tau_{yx}$$

$$\overset{(n_y)}{T}_y = \lim_{\Delta A_y \rightarrow 0} \frac{\Delta F_y}{\Delta A_y} = \sigma_{yy}$$

$$\overset{(n_y)}{T}_z = \lim_{\Delta A_y \rightarrow 0} \frac{\Delta F_z}{\Delta A_y} = \tau_{yz}$$

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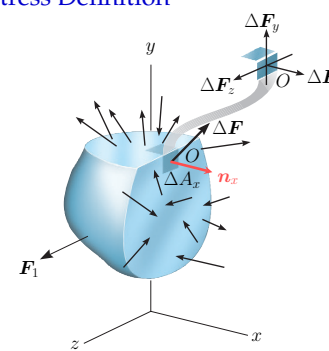
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Concept of Stress

Stress Definition



- Stress vector at point O is :

$$\overset{(n_x)}{T} = \lim_{\Delta A_x \rightarrow 0} \frac{\Delta F}{\Delta A_x}$$

- Three component of stress vector on face n_x at O is:

$$\overset{(n_x)}{T}_x = \lim_{\Delta A_x \rightarrow 0} \frac{\Delta F_x}{\Delta A_x} = \sigma_{xx}$$

$$\overset{(n_x)}{T}_y = \lim_{\Delta A_x \rightarrow 0} \frac{\Delta F_y}{\Delta A_x} = \tau_{xy}$$

$$\overset{(n_x)}{T}_z = \lim_{\Delta A_x \rightarrow 0} \frac{\Delta F_z}{\Delta A_x} = \tau_{xz}$$

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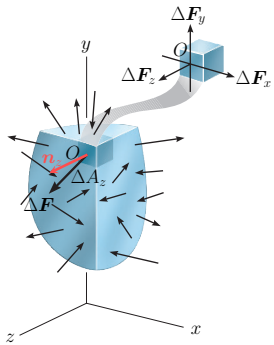
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Concept of Stress

Stress Definition



- Stress vector at point O is :

$$\mathbf{T}^{(n_z)} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta \mathbf{F}}{\Delta A_z}$$

- Three component of stress vector on face n_z at O is:

$$T_x^{(n_z)} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_x}{\Delta A_z} = \tau_{zx}$$

$$T_y^{(n_z)} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_y}{\Delta A_z} = \tau_{zy}$$

$$T_z^{(n_z)} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_z}{\Delta A_z} = \sigma_{zz}$$

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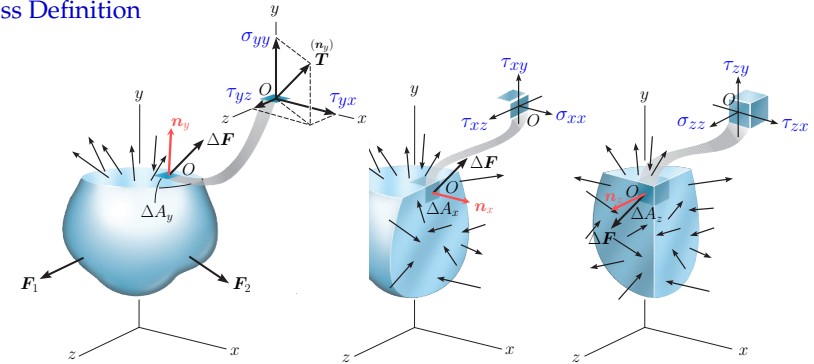
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Concept of Stress

Stress Definition



- σ_{ij} - indicates the component of stress acting on a face with normal i and in the direction j

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Concept of Stress

Q What is the direction of stress or traction vector?

- Static equilibrium: $\sum F = 0, \sum M = 0$
- Moment equilibrium translates to:

$$\tau_{xy} = \tau_{yx}; \quad \tau_{yz} = \tau_{zy}; \quad \tau_{zx} = \tau_{xz}$$

There are only six unique components of stress required to define the state of stress at a given point.

- The shear cannot take place in one plane only. An equal shear stress must be extended on another plane perpendicular to the first one.
- The normal and shear stresses can be defined as:

$$\sigma_N = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_N}{\Delta A}; \quad \sigma_S = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_S}{\Delta A}$$

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Concept of Stress

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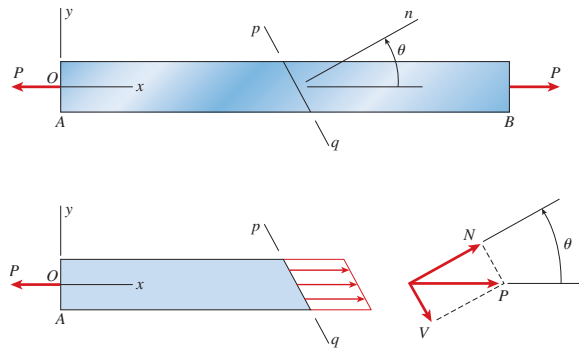
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Axial Loading

Stresses on an oblique plane



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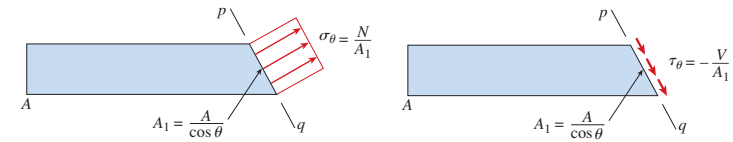
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Axial Loading

Stresses on an oblique plane



- Under uniaxial loading :

$$F = N = P \cos \theta; \quad V = P \sin \theta; \quad A_1 = \frac{A}{\cos \theta}$$

$$\sigma_\theta = \frac{F}{A_1} = \frac{P}{A} \cos^2 \theta; \quad \tau_\theta = -\frac{V}{A_1} = -\frac{P}{A} \sin \theta \cos \theta$$

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Axial Loading

[Example 1] What are the maximum normal and shear stresses under uniaxial loading?

Axial Loading

[Example 2] Under uniaxial loading, the maximum allowable shear stress is 90 MPa, $A = 18 \text{ mm}^2$, $\theta = 50^\circ$.

- 1 Determine the largest load P that can be applied?
 - 2 What is the corresponding tensile stress?
- Step 1: The shear stress under uniaxial loading is:

$$\tau = \frac{P}{A} \cos \theta \sin \theta = \frac{P}{18} \cos 50^\circ \sin 50^\circ = 0.0273P \implies P = \frac{\tau}{0.0273} = \frac{90}{0.0273} = 3290 \text{ N}$$

- Step 2: The normal stress, therefore, is:

$$\sigma = \frac{P}{A} \cos^2 \theta = \frac{3290}{18} \cos^2 50^\circ = 75.5 \text{ N/m}^2$$

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Stress Tensor

Q How is the traction vector related to the second order stress tensor?

Consider a traction vector acting on any arbitrary plane with a normal with direction cosines (l, m, n) . The components of the traction vector are linear combination of the direction cosines with respective components of the stress tensor. Let us prove it !

Reading Assignment

Chapter 2: Section 2.1 - 2.3.1